

SYNOPSIS

Report On

FinTrack - A Finance Tracker

By

Nitin Gangwar - 202410116100136

Neha Agnihotri – 202410116100131

Paras Chandravanshi - 202410116100140

Rishabh Singh Bhadauriya-202410116100166

Session:2024-2025 (IVth Semester)

Under the supervision of

Dr.Amit Kumar

Asistant Professor

KIET Group of Institutions, Delhi-NCR, Ghaziabad



**DEPARTMENT OF COMPUTER APPLICATIONS
KIET GROUP OF INSTITUTIONS, DELHI-NCR,
GHAZIABAD-201206
(2025 - 2026)**

ABSTRACT

The **FinTrack – Finance Tracker** is a web-based financial management application designed to help users efficiently track, manage, and analyze their daily income and expenses. In today's fast-paced digital world, managing personal finances manually can be time-consuming and prone to errors. FinTrack provides a simple, secure, and user-friendly platform that enables users to monitor their financial activities in real time.

The application is developed using **Python** for the frontend to ensure a responsive and dynamic user interface, with Express.js for the backend to handle server-side logic and API management, and as the database to store and manage user financial data securely. The system follows a RESTful architecture for smooth communication between client and server.

FinTrack allows users to register and log in securely, add income and expense records, categorize transactions, edit or delete entries, and view summarized reports. The application also provides graphical representations of financial data to help users analyze spending patterns and make informed financial decisions.

TABLE OF CONTENTS

	Page Number
1. Introduction	4
2. Literature Review	5
3. Project Objective	6
4. Hardware and Software Requirements	7
5. Project Flow	8
6. Project Outcome	9
7. Proposed Time Duration	9

INTRODUCTION

In today's digital era, managing personal finances efficiently has become essential for individuals to maintain financial stability and achieve their future goals. Many people face difficulties in tracking their daily income and expenses, which often leads to poor budgeting and unnecessary financial stress. Traditional methods such as maintaining handwritten records or spreadsheets can be time-consuming, error-prone, and inconvenient.

To address these challenges, **FinTrack – Finance Tracker** is developed as a modern web-based application that helps users systematically manage their financial activities. The project aims to provide a simple, secure, and interactive platform where users can record their income and expenses, categorize transactions, and monitor their financial performance in real time.

FinTrack is built using modern web technologies including **Python** for creating a responsive and dynamic user interface, with Express.js for backend server operations, and for efficient data storage and management. The application follows a RESTful architecture to ensure smooth communication between frontend and backend systems.

LITERATURE REVIEW

Overview of Personal Finance Management Systems

Personal finance management has evolved significantly from traditional manual methods to modern digital platforms. Earlier, individuals maintained records of income and expenses using notebooks, registers, or spreadsheets. While these methods were simple, they were prone to errors, lacked automation, and did not provide analytical insights. With the advancement of technology, web-based and mobile applications have replaced manual systems, offering better accuracy, efficiency, and convenience.

Existing Finance Tracking Applications

Several financial management applications have been developed to assist users in budgeting and expense tracking. Popular applications such as **Mint**, **YNAB (You Need A Budget)**, and **PocketGuard** provide features like automated expense categorization, financial goal setting, real-time tracking, and graphical reports.

These systems demonstrate the effectiveness of digital platforms in promoting financial awareness. However, many of these applications have limitations such as subscription costs, complex interfaces for beginners, limited customization options, and privacy concerns.

Security Considerations in Finance Systems

Financial applications require strong security mechanisms to protect sensitive user data. Research highlights the importance of:

- Secure user authentication and authorization
- Password encryption techniques
- Token-based authentication (JWT)
- Secure API routes
- Data validation and error handling

Security is a critical factor in gaining user trust and ensuring data integrity.

Need for a Customized Finance Tracker

Although many finance tracking applications exist, there is still a need for a simple, customizable, and user-friendly system that caters to individual requirements without unnecessary complexity. Many existing platforms focus on advanced financial tools, which may not be suitable for users who only need basic income and expense tracking.

FinTrack – Finance Tracker is developed to bridge this gap by providing a lightweight, secure, and scalable web-based solution. It focuses on simplicity, efficiency, and real-time financial monitoring using modern technologies such as React.js, Node.js, and MongoDB.

PROJECT OBJECTIVES

The main objective of **FinTrack – Finance Tracker** is to develop a secure, user-friendly, and efficient web-based application that helps users manage and monitor their personal financial activities effectively.

- **User Authentication System**

- To implement secure user registration and login functionality.
- To ensure data privacy through authentication and authorization mechanisms.

- **Income and Expense Management**

- To allow users to add, edit, and delete income and expense records.
- To categorize transactions for better organization and tracking.

- **Real-Time Financial Tracking**

- To provide instant updates of total income, total expenses, and current balance.
- To display financial summaries dynamically.

- **Data Visualization and Reports**

- To generate graphical representations (charts/graphs) for analyzing spending patterns.
- To help users make informed financial decisions based on reports.

- **Responsive and User-Friendly Interface**

- To design an intuitive and interactive UI using React.js.
- To ensure compatibility across desktop and mobile devices.

- **Secure Database Management**

- To store financial data securely using MongoDB.
- To maintain data integrity and reliability.

- **Scalability and Performance**

- To develop the system using RESTful architecture for smooth communication between frontend and backend.
- To ensure the application can handle multiple users efficiently.

HARDWARE AND SOFTWARE REQUIREMENTS

This section lists the technical requirements needed for developing and running the application.

- **Hardware Requirements:**

- **Processor:** Dual-core CPU (e.g., Intel i3 or equivalent) or better for desktop/laptop At least 2GB RAM and 500MB storage space for application and data.
- **Memory (RAM):** 4 GB or higher recommended for smooth use.
- **Network:** Stable broadband internet connectivity (≥ 1 Mbps).

- **Software Requirements:**

- **Backend Framework:** Python with Streamlit, Cohere API, Pandas, Plotly
- **Database:** SQLite Database or similar relational/NoSQL DB.
- **Web Server:** Streamlit or FastApi
- **Development Tools:** Visual Studio Code for coding, Git for version control, and a web browser for testing.

PROJECT FLOW

The project flow explains how data moves from the user interface to the database and back to the user.

1. User Interaction Layer:

- User registers or logs into the system.
- User enters income or expense details.
- User views dashboard, reports, and analytics.
- React.js handles UI rendering and form validation.
- React.js handles UI rendering and form validation.

2. API Communication Layer:

- Frontend sends HTTP requests (GET, POST, PUT, DELETE).
- Data is transferred in JSON format.

3. Business Logic Layer:

- Server receives client request
- Authentication middleware verifies JWT token.

4. Database Layer:

- User details stored securely
- Transactions stored in collections.
- Data retrieved, updated, or deleted based on request.
- Passwords stored in encrypted format (bcrypt).

5. Response & UI Update:

- Backend sends JSON response to frontend.
- React updates dashboard dynamically.
- Charts and reports reflect updated financial data.

6. Authentication Flow:

- User registers → Password encrypted.
- User logs in → JWT token generated.
- Token stored in browser.
- Protected routes require valid authentication.

7. Development Flow:

- Requirement Analysis
- System Design
- Frontend Development.
- Backend API Development.
- Database Design.
- Integration & Testing.
- Deployment.

PROJECT OUTCOMES

Expected outcomes of FinTrack-Finannce Tracker include:

- **Functional Outcome:**
A Secure user registration and login system.
Efficient income and expense management.
Automatic calculation of total income, expenses, and balance.
Real-time dashboard updates.
Graphical representation of financial data.
- **Technical Outcome:**
Successful implementation of MERN stack architecture.
Development of RESTful APIs.
Secure JWT-based authentication system.
Efficient database management using MongoDB.
Responsive and user-friendly interface.
- **User Level Outcome:**
Helps users track daily financial activities.
Provides better financial planning and budgeting.
Reduces paperwork and manual calculations.
Encourages financial awareness and savings habits.
- **Academic Outcome :**
Encourages financial awareness and savings habits.
Practical experience in client-server architecture.
Improved database management skills.

PROPOSED TIME DURATION

This section outlines the timeline for project development, including milestones:

- **Weeks 1-2:** Requirement gathering and initial design of the application.
- **Weeks 3-4:** Development of the backend, including database setup.
- **Weeks 5-6:** Frontend development, user interface design, and integration with the backend.
- **Weeks 7-8:** Testing, debugging, and final deployment of the application.

